

Voron Pi Update Walkthrough

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What this is and why

The Voron's Raspberry Pi needs four updates, done in a fixed order: package sources, operating system, printer software, then the main board's firmware. Nothing here changes the printer's settings (`printer.cfg` and friends stay as they are).

Why now:

- **The Pi's operating system is out of support.** It runs Debian 11 "Bullseye", which reached end of life on August 31, 2026. Debian moved its packages to an archive server, so `apt upgrade` now fails with hundreds of 404 errors.
- **The main board's firmware is about three years old.** The BTT Octopus runs Klipper v0.11.0 firmware, while the Pi now runs Klipper v0.13. Mainsail shows a warning: *MCU 'mcu' has deprecated code... Recompiling and flashing is recommended.* It still prints, but the gap will eventually become a hard error.

What it touches: the Pi's package list, the Klipper/Moonraker/Mainsail software, and the Octopus firmware. The Beacon probe updates its own firmware and needs nothing by hand.

What it does not do: reinstall the operating system. That's a separate, later job (see the last section).

Risk: low. The config gets backed up first, and the firmware flash goes through the Octopus's built-in bootloader, which the flash never overwrites. A wrong setting means the board won't connect until it's reflashed correctly. It does not brick the board.

The order, and why it matters

Each step depends on the one before it. Don't skip ahead.

1. **Back up the config.** Insurance for everything that follows.
2. **Fix the package sources.** `apt` can't download anything until it points at Debian's archive server.
3. **Upgrade the OS packages, then reboot.** Klipper and Moonraker run on top of these libraries (Python, systemd, OpenSSL), so the foundation goes first.

4. **Update the printer software in Mainsail's update manager.** Order: Moonraker, then Klipper and Beacon together, then everything else. Klipper and the Beacon plugin must end up on matching versions, because the plugin hooks into Klipper's Z homing code, which changed in May 2026.
5. **Reflash the Octopus firmware.** It has to be built from the Klipper version installed in step 4. Flash it any earlier and it will be out of date again as soon as Klipper updates.
6. **Verify.** Restart, check versions, home, level, and read the logs.

If update manager was already run before steps 2 and 3, that's fine. Do steps 2 and 3, then open update manager again and confirm everything shows as up to date before moving to step 5.

Before you start

You'll need:

- A computer that can SSH into the Pi (user `pi`). Replace `<printer-ip>` below with the Pi's address.
- A microSD card, 32 GB or smaller, formatted FAT32, and a way to write to it.
- Physical access to the printer. The Octopus's SD slot is inside the electronics bay.
- No print running.

Back up the config (step 1). On the Pi:

```
cd ~
tar czf printer_data-backup-$(date +%F).tgz printer_data/config
```

Then copy it off the Pi, from your own computer:

```
scp pi@<printer-ip>:~/printer_data-backup-*.tgz ~/Desktop/
```

Note the current Klipper version in case you need to roll back:

```
git -C ~/klipper log -1 --format='%h %cs'
```

Write it here: _____

Step 2: Point apt at Debian's archive

Save a copy of the current sources file, then look at what's there:

```
sudo cp /etc/apt/sources.list /etc/apt/sources.list.bak
cat /etc/apt/sources.list
ls /etc/apt/sources.list.d/
```

Edit the file with `sudo nano /etc/apt/sources.list`. Replace every Debian line (`deb.debian.org` or `security.debian.org`) with these two:

```
deb http://archive.debian.org/debian bullseye main contrib non-free
deb http://archive.debian.org/debian-security bullseye-security main contrib
non-free
```

- Delete any `bullseye-updates` or `bullseye-backports` line, or put `#` in front of it.
- Leave `raspi.list` in `/etc/apt/sources.list.d/` alone. That's the Raspberry Pi repo, and it still works.

Save with Ctrl+O, Enter, then exit with Ctrl+X. Then:

```
sudo apt update
```

Expected: no 404 errors. If it says a Release file "is not valid yet" or "has expired", run this instead:

```
sudo apt -o Acquire::Check-Valid-Until=false update
```

Step 3: Upgrade the OS packages and reboot

```
sudo apt upgrade
```

- Answer **Y** to continue.
- If it asks whether to keep or replace a config file (nginx, sshd), choose **keep your currently-installed version**. That's the default, so just press Enter.
- About 125 packages should install, the same list that failed before (systemd, OpenSSL, Python 3.9, nginx and so on).

When it finishes:

```
sudo reboot
```

Wait for the Pi to come back, then confirm Mainsail loads in the browser before going on.

Step 4: Update the printer software

4a. Check the Klipper folder for local edits first. Klipper's version currently ends in `-dirty`, which means files in `~/klipper` were edited. Update manager may refuse to update a dirty repo.

```
git -C ~/klipper status --short
git -C ~/klipper diff --stat
```

- **Nothing listed:** go to 4b.
- **Files listed, but `diff --stat` shows 0 insertions, 0 deletions:** only file permissions changed. Fix it with `git -C ~/klipper config core.fileMode false` and check again.
- **Real line changes:** stop. Don't use "Hard Recovery" in Mainsail, because that wipes the changes. Save the `git diff` output and decide whether those edits matter before continuing.

4b. Run update manager in this order. In Mainsail, go to **Machine → Update Manager** and click **Refresh**. Then:

1. **System** (OS packages). Should now show nothing to do after step 3.
2. **moonraker**
3. **klipper**, then **beacon** right after it. Never leave one updated and the other behind.
4. Everything else: **mainsail**, **mainsail-config**, **KlipperScreen**, **crowsnest**, **led_effect**, **sonar**, **moonraker-telegram-bot**.

Expected: every row shows as up to date, none as *dirty* or *invalid*.

Note: Moonraker 0.11 now rejects a wrong API key instead of falling back to trusted-network access. If the Telegram bot stops responding afterwards, that's the likely cause. It's a nice-to-have, not needed to print.

Step 5: Reflash the Octopus firmware

The Octopus (STM32F446) talks to the Pi over the Pi's GPIO serial pins (`/dev/ttyAMA0`), not USB. So the firmware goes on by SD card.

5a. Build the firmware on the Pi:

```
cd ~/klipper
make clean
make menuconfig
```

The menu opens with leftover settings for a different chip (atmega32u4). Change **every** line to match this table:

Setting	Value
Enable extra low-level configuration options	Checked
Micro-controller Architecture	STMicroelectronics STM32
Processor model	STM32F446
Bootloader offset	32KiB bootloader
Clock Reference	12 MHz crystal
Communication interface	Serial (on USART2 PD6/PD5)
Baud rate for serial port	250000

The Communication interface line matters most. Pick USB by mistake and the board will flash but the Pi can't reach it.

Press **Q**, then **Y** to save. Then:

```
make
ls -l out/klipper.bin
```

Expected: `out/klipper.bin` exists.

5b. Copy it to the microSD card. From your own computer:

```
scp pi@<printer-ip>:~/klipper/out/klipper.bin ~/Desktop/firmware.bin
```

Copy `firmware.bin` to the root of the FAT32 card. The name must be exactly `firmware.bin`.

5c. Flash:

1. Power the printer off at the main switch.
2. Insert the card in the Octopus's microSD slot.

3. Power on and wait about 30 seconds.
4. Power off and remove the card.
5. Check the card on your computer. If the file is now named `FIRMWARE.CUR`, the flash worked.
6. Power the printer back on.

The Beacon probe updates its own firmware when Klipper starts, so there's nothing to do for it. If the ADXL accelerometer (rp2040) is plugged in and Mainsail shows a version warning for it too, it needs its own build. It isn't needed for printing, so leave it unplugged for now.

Step 6: Verify

In Mainsail, with the printer powered on:

- ☐ **No MCU warning.** The *deprecated code* message is gone.
- ☐ **Versions match.** Under Machine → System Loads, `mcu` shows a v0.13.0 version that matches the Klipper host version, and `beacon` shows its version.
- ☐ **Home all (G28).** All three axes home normally. Watch the first Z home closely, with a hand near the power switch.
- ☐ **Quad gantry level (QUAD_GANTRY_LEVEL).** Completes without errors.
- ☐ **Heaters.** Set the bed to 60 °C and the nozzle to 150 °C. Both reach temperature and hold it.
- ☐ **Logs are clean.** On the Pi:

```
grep -iE "deprecat|error|mismatch" ~/printer_data/logs/klippy.log | tail -30
grep -iE "deprecat|error|warning" ~/printer_data/logs/moonraker.log | tail -30
```

Some old error lines from before the update are normal. Look at the newest ones, at the bottom.

If something goes wrong

Symptom	Likely cause	Fix
<code>apt update</code> still shows 404s	An old Debian line is still active	Recheck <code>/etc/apt/sources.list</code> and the files in <code>/etc/apt/sources.list.d/</code>
"Release file expired" or "not valid yet"	Archived releases carry old dates	Use the <code>-o Acquire::Check-Valid-Until=false</code> form from step 2
Update manager shows Klipper as <i>dirty</i>	Local edits in <code>~/klipper</code>	Step 4a. Don't click Hard Recovery until the changes have been checked
"mcu: Unable to connect" after flashing	Wrong Communication interface in menuconfig	Rebuild with Serial (on USART2 PD6/PD5) and flash again. The bootloader is untouched
<code>firmware.bin</code> not renamed to <code>FIRMWARE.CUR</code>	Card not FAT32, card too large, or wrong file name	Use a 32 GB or smaller FAT32 card and name the file exactly <code>firmware.bin</code>
Homing or QGL acts strangely	Klipper and the Beacon plugin out of step	Update manager: make sure beacon is current, then <code>FIRMWARE_RESTART</code>

Restore the config if a file got changed by mistake:

```
cd ~
tar xzf printer_data-backup-YYYY-MM-DD.tgz
sudo systemctl restart klipper
```

Roll Klipper back (last resort), using the version written down in Before you start:

```
git -C ~/klipper checkout <version-you-wrote-down>
sudo systemctl restart klipper
```

Then rebuild and reflash the Octopus (step 5) from that version, so the board and the Pi match again.

Later: a fresh operating system

These steps get apt working again, but Bullseye won't receive any more security fixes. That's acceptable for a printer on a home network. Don't expose Mainsail to the internet (no port forwarding).

The proper fix is to reflash the Pi's SD card with current Raspberry Pi OS Lite (Bookworm or Trixie). Then reinstall Klipper, Moonraker, Mainsail, KlipperScreen and Beacon with KIAUH, and copy `printer_data/config` back in. Plan it after the first-layer work is settled, so the OS and the printer config don't change at the same time.

Sign-off

- ☐ Step 1: Config backed up and copied off the Pi
- ☐ Step 2: apt sources point at archive.debian.org; `apt update` runs clean
- ☐ Step 3: OS packages upgraded; Pi rebooted
- ☐ Step 4: Update manager all current (Moonraker → Klipper + Beacon → the rest)
- ☐ Step 5: Octopus flashed; `FIRMWARE.CUR` on the card
- ☐ Step 6: No MCU warning; homing, QGL and heaters OK; logs clean

Approved to proceed: _____ Date: _____

Done by: _____ Date: _____